Speaking App PRD — Student Form

- **최정환 / 2025315143**
- **My situation (상황):** Practicing a sincere apology, understanding a parent's intention, and expressing one's own feelings calmly during a family misunderstanding.
  Example: The learner comes home tired and takes a short break on the phone. Mom keeps asking about homework because she is worried, and the learner answers too sharply.

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1. App Name (앱 이름)

✏️ **My answer:**

- **Talk It Out! — Family Talk**

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2. Why We Are Building It (왜 만드는가)

✏️ **My answer:**

- **Problem:** Students may know how to say "sorry," but they may react with irritation when they feel tired or misunderstood. They need practice apologizing, explaining their feelings, and listening to the other person.
- **Solution:** Students practice a five-turn conversation with Mom through **Listen → Speak → Get feedback → Try again**. Each successful sentence rebuilds one part of a bridge between them.
- **Goal:** Students can say five simple expressions for apologizing, understanding another person, expressing feelings, and finding a solution intelligibly.

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3. Target Users (대상 사용자)

✏️ **My answer:**

- **Grade:** Middle school 2nd year
- **Level:** CEFR A2
- **Characteristic:** Students can use basic English but may feel uncomfortable talking about feelings or conflict.
- **Common mistake:** Leaving out short words such as *to* or *you* when speaking quickly.
- **Use context:** On a smartphone, for 5–10 minutes, during class review or homework.

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4. Screen Layout (화면 구성)

🖍️ **My answer:**

Screen 1: Start Screen

- **Top:** App title **Talk It Out! — Family Talk** + short situation: "You came home tired and were taking a short break. Mom asked about your homework again, and you answered too sharply."
- **Middle:** Learner and Mom on opposite sides of a broken bridge + Mom's message: **"I was only worried about you."**
- **Bottom:** [Start Conversation] + privacy notice

Screen 2: Practice Screen

- **Top:** Mom's sentence + Korean meaning + **Bridge 3/5**
- **Middle:** Student target sentence in the **largest text** + [🔊 Listen] [🐢 Slow]
- **Bottom:** Large [🎤 Mic] + recognized speech + feedback + [Try Again] [Next →] [Skip]
- **Mic button states: Ready → Listening → Done**

Screen 3: Result Screen

- **Top: "The bridge is complete!"** or **"You calmly finished the conversation!"**
- **Middle:** Learner and Mom talking calmly + **Apologize → Explain → Understand → Express → Compromise**
- **Bottom:** Completion stars + **"Understanding starts when both sides speak and listen."** + [Practice Again] [Home]

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5. Key Features (주요 기능)

🖍️ **My answer:**

1. **See the situation** — Mom's sentence appears and is read aloud once.
2. **Listen** — [🔊 Listen] plays the target sentence; [🐢 Slow] plays it slowly.
3. **Speak** — Tap [🎤 Mic] and say the target sentence.
4. **Compare** — Compare the recognized words with the target sentence. Ignore capitalization and punctuation. If all key words are recognized, it counts as success.
5. **Feedback** — Mom gives short feedback. A successful sentence rebuilds one bridge piece.
6. **Try again / Next** — Students can retry, move to the next turn, or skip. Retrying has no penalty.

After all five turns, the app moves to the Result Screen.

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6. Practice Dialogue (연습 대화)

🖍️ **My answer:**

- **Partner character (name / role):** Mom / the learner's parent

Situation

The learner comes home tired after school and academy and takes a short break on the phone. Mom asks about homework because the learner has been sleeping late and looking tired. The learner feels pressured and answers sharply. The learner should apologize for the harsh tone, explain the situation, understand Mom's concern, express personal feelings calmly, and suggest a solution.

No.	Partner	Student target sentence	Korean meaning	Key words
1	I was only asking about your homework. You sounded angry.	**I'm sorry I spoke like that.**	그렇게 말해서 미안해.	sorry, spoke
2	I thought you were just playing on your phone.	**I was just taking a short break.**	나는 그냥 잠깐 쉬고 있었어.	taking, break
3	I asked because you've been tired lately.	**I know you were worried about me.**	엄마가 나를 걱정한 거 알아.	know, worried
4	I just want you to finish your homework.	**I understand, but I feel stressed when you keep asking.**	이해해. 하지만 계속 물어보면 스트레스를 받아.	understand, stressed, asking
5	Okay. What would help?	**Can I rest for ten minutes, then do my homework?**	10분 쉬고 나서 숙제해도 될까?	rest, ten minutes, homework

- **Star rules (reward completion, not accuracy):**
  - ⭐⭐⭐ Finished all 5 turns without [Skip]. Retrying is allowed.
  - ⭐⭐ Finished with 1 skipped turn.
  - ⭐ Finished the whole dialogue with 2 or more skipped turns.

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7. Feedback Rules (피드백 규칙)

🖍️ **My answer:**

- **Character personality / speaking style:** Mom is caring and worried, but her repeated questions may feel like pressure. She becomes calmer as the learner explains their feelings. She uses short CEFR A2 English and focuses on meaning, not accent.
- **My rules:**
    1. Praise first and correct only **one** thing at a time.
    2. Use **two sentences or fewer**, with simple English + Korean.
    3. Judge by **"Did Mom understand?"**, not native-like pronunciation.
    4. No "wrong," percentage scores, or mouth/tongue explanations.
    5. Use **1st recast → 2nd elicitation → 3rd chunking**.

Situation	Feedback (English + Korean)
Success	**Thanks for telling me calmly. I understand.** (차분하게 말해 줘서 고마워. 이해했어.)
Success + small word missing	**Good! I understood you. Try adding the small word next time.** (뜻은 잘 전달됐어요. 다음에는 빠진 짧은 단어도 넣어 보세요.)
1st failure (recast)	**Nice try! You can say, "{current target sentence}."** (좋아요! 이 문장으로 다시 말해 볼까요?)
2nd failure (elicitation)	**Almost! Try the missing word: "{first letter}___."** (거의 다 왔어요! 첫 글자를 보고 다시 말해 보세요.)
3rd failure (chunking)	**Good effort! Say it in parts: "{chunk 1}" / "{chunk 2}."** (잘했어요! 문장을 나누어 말해 봐요.)
No speech recognized	**That's okay! Tap the mic and try again.** (괜찮아요! 마이크를 누르고 다시 말해 봐요.)

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8. Do Not (하지 말 것)

🖍️ **My answer:**

Privacy and Safety (required — keep these)

- Do not save or send audio recordings.
- Do not collect personal information (name, student ID, school, email). No login or sign-up.
- No ads, external links, or payment features.
- No photos of real people or famous characters.

Learning Design

- Do not give pronunciation percentages or native-speaker comparison scores.
- Do not change or create new target sentences.
- Do not make the learner apologize for having feelings or needing rest. The learner apologizes only for speaking harshly.
- Do not present Mom as always right or the learner as simply disobedient.
- Do not punish retries or speaking mistakes.

Other

- Do not use childish animations, shouting effects, or dramatic family conflict.

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9. Completion Checklist (완성 체크리스트)

🖍️ **My answer:**

- ☐ The mic button changes to **“Listening”** when tapped.
- ☐ What the student said appears as text.
- ☐ Capitalization and punctuation differences are not counted as failures.
- ☐ Saying nothing shows a **“Let's try again”** message.
- ☐ Each Practice Screen shows Mom’s sentence and the target sentence.
- ☐ The target sentence is the largest text on the Practice Screen.
- ☐ Recognizing all key words rebuilds one bridge piece.
- ☐ Failed attempts show **recast → elicitation → chunking** in order.
- ☐ After all five turns, the Result Screen appears.
- ☐ Stars are based on completion and skipped turns, not pronunciation accuracy.

Technical Requirements (do not edit — 수정하지 말고 그대로 두세요)

- **Speech recognition:** Browser built-in Web Speech API (SpeechRecognition), language `en-US`
- **Read-aloud:** Browser built-in SpeechSynthesis, language `en-US`, normal rate 0.9 / slow rate 0.7
- **Feedback sentences:** Gemini, following Section 7. If Gemini does not respond, show the matching example sentence from the Section 7 table. The app must work to the end without AI.
- Do **not** run SpeechRecognition and MediaRecorder at the same time. Use speech recognition only.
- Disable the mic button while the model audio is playing.
- Show a guidance message when: the browser does not support speech recognition ("Please open it in Chrome."), microphone permission is denied, or the app is opened in an in-app browser such as KakaoTalk or Instagram.
- Show a small notice on the Start Screen: "Your voice is processed to turn it into text and is not recorded or saved."
- The deployed address must use **https** for the microphone to work.

Before You Submit (제출 전 자기 점검)

- [ ] Section 3 includes a **CEFR level**, and the sentences in Section 6 match it.
- [ ] Section 4 uses **Top / Middle / Bottom** and matches my Stitch screens.
- [ ] Section 6 has **5 turns** with key words.
- [ ] Section 7 has **recast → elicitation → chunking** and no % scores.
- [ ] Section 9 has sentences I can check by testing.
- [ ] My partner checked this PRD, and I revised it.
- [ ] I recorded the AI tools and main prompts in my AI use log (.md).